

Sub-Semigroups & Submonoids:-

Let $(S, *)$ be semigroup and $T \subseteq S$. If T is closed under the operation $*$, then $(T, *)$ is said to be Sub semigroup of $(S, *)$

Let $(M, *, e)$ be a monoid and $T \subseteq M$. If T is closed under the operation $*$ and $e \in T$, then $(T, *, e)$ is said to be a submonoid of $(M, *, e)$

Eg: For the semigroup $(\mathbb{N}, +)$, T - set of all multiples of a +ve integer 'm', then $(T, +)$ is subsemigroup of $(\mathbb{N}, +)$

$$T = \{0, 3, 6, 9, 12, \dots\}$$

ii) For $(\mathbb{N}, +)$, Set of all even non negative integers $\in E = \{0, 2, 4, 6, 8, \dots\}$

iii) For $(\mathbb{I}, +)$, Set of multiples of an integer 'm' $\rightarrow T$

$$T = \{\dots, -8, -4, 0, 4, 8, \dots\}$$

$(T, +)$ is submonoid of $(\mathbb{I}, +)$

iv) For $(\mathbb{I}, \times)$, $T = \{\text{set of multiples of an integer 'm'}\}$

$$T = \{\dots, -9, -6, -3, 0, 3, 6, 9, \dots\}$$

$e \notin T \therefore T$ is not submonoid.

v) for $(\mathbb{Z}_m, \times_m)$ for $m=5$ & 6 - monoids.

$\{[1], [2], [3], [4]\}$ and $\{[1], [5]\}$ are submonoids.

Thm For any commutative monoid $(M, *, e)$ the set of idempotent elts of M forms a submonoid.

Let $S = \{\text{all idempotent elts of } M\}$

take $a, b \in S$. $\times: S \times S \rightarrow S$
 $\times(a, b) \rightarrow a \times b$

$$\Rightarrow a \times a = a \text{ \& } b \times b = b$$

$$\nexists (a \times b) \times (a \times b) = a \times b$$

$$\begin{aligned} (a \times b) \times (a \times b) &= (a \times b) \times (b \times a) \quad \text{M is commutative.} \\ &= a \times b \times b \times a \\ &= a \times (b \times b) \times a \quad \text{b is idempotent} \\ &= a \times b \times a \\ &= a \times a \times b \quad \text{Commutativity} \\ &= a \times b \end{aligned}$$